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Homework #6

习题五 T2, T7, T12, T14, T16

T2 $X(t)=A\sin(\omega t+\Theta)$ $\Theta \sim -\frac{\pi}{2}$ $\frac{\pi}{2}$ $A\sim U(-1,1)$

P 步 主

$\mu_X(t)=E[X(t)]=E[A]E[\sin(\omega t+\Theta)]=0$ 为常数

$R_X(t,t+\tau)=E[X(t)X(t+\tau)]$

$=E[A^2]E[\frac{\cos\tau-\cos(2\omega t+\tau+2\Theta)}{2}]$

$=\frac{1}{6}\cos\tau$ 只与τ有关

故$\{X(t),-\infty<t<\infty\}$为平稳过程

T7 $X(t)=B(t+1)-B(t)\sim N(0,1)$

(1) $\mu_X(t)=E[B(t+1)-B(t)]=0$

$R_X(t,t+\tau)=E[X(t)X(t+\tau)]=\begin{cases}1-|\tau| & |\tau|<1\\ 0 & \text{其他}\end{cases}$

(2) $\mu_X(t)$为常数，$R_X(t,t+\tau)$只与τ有关

故$X(t)$为宽平稳过程

又$X(t)$为正态过程 故$X(t)$为严平稳过程

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No. $\int_{-T}^{T}\cos^2 t\,dt = T+\frac{1}{2}\sin 2T$ $\int_{-T}^{T}\sin^2 t\,dt = T-\frac{1}{2}\sin 2T$

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$X(t) = \sqrt{2}X\cos t + Y\sin t$ $f(x) = \begin{cases} 1-|x| & -1<x<1 \\ 0 & \text{其他} \end{cases}$

$Y \sim U(-1,1)$

(1) $\mu_X(t) = E[X(t)] = 0$ 为常数

$R_X(t, t+\tau) = E[X(t)X(t+\tau)]$

$E[X^2] = \int_{-1}^{1} x^2(1-|x|)\,dx = \frac{1}{6}$ $E[Y^2] = \frac{1}{3}$

$E[XY] = \int_{-1}^{1}\int_{-1}^{1} xy(1-|x|)\,dx\,dy = 0$

$R_X(t, t+\tau) = \frac{1}{3}\cos t\cos(t+\tau) + \frac{1}{3}\sin t\sin(t+\tau)$

$= \frac{1}{3}\cos\tau$ 仅与 τ 有关

故 $\{X(t); -\infty<t<\infty\}$ 是平稳过程。

(2) $\langle X(t)\rangle = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} X(t)\,dt$

$= \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}(\sqrt{2}X\cos t + Y\sin t)\,dt$

$= \lim_{T\to\infty}\frac{\sqrt{2}X\sin T}{T} = 0 = \mu_X(t)$

均值具有各态历经性

(3) $\langle X(t)X(t+\tau)\rangle = \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T} X(t)X(t+\tau)\,dt$

$= \lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}[2X^2\cos t\cos(t+\tau) + Y^2\sin t\sin(t+\tau)]$

$= (X^2+\frac{1}{2}Y^2)\cos\tau \neq R_X(t, t+\tau)$ 非各态历经过程

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$P\{X=k\} = \frac{e^{-\lambda}\lambda^k}{k!}$ $E[X^2] = E(X)^2 + D(X) = \lambda^2 + \lambda$ ……

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T14 $X(t) = A[N(t+1) - N(t)]$ $A \sim U(0,1)$, $N(t+1) - N(t) \sim \pi(1)$

(1) $\mu_X(t) = E[A]E[N(t+1) - N(t)] = \frac{1}{2}$

$R_X(t, t+\tau) = E[A^2]E\{[N(t+1) - N(t)][N(t+1+\tau) - N(t+\tau)]\}$

$E[A^2] = \int_0^1 x^2 dx = \frac{1}{3}$

当 $|\tau| > 1$ 时, $R_X(t, t+\tau) = \frac{1}{3}$

当 $0 < \tau < 1$ 时, $E\{[N(t+1) - N(t)][N(t+1+\tau) - N(t+\tau)]\}$

$= E\{[N(t+1) - N(t+\tau) + N(t+\tau) - N(t)][N(t+1+\tau) - N(t+\tau) + N(t+\tau) - N(t+\tau)]\}$

$= 2 - \tau$

同理当 $-1 < \tau < 0$ 时, $E\{[N(t+1) - N(t)][N(t+1+\tau) - N(t+\tau)]\} = 2 + \tau$

故 $R_X(t, t+\tau) = \begin{cases} 2 - |\tau| & |\tau| < 1 \\ \frac{1}{3} & |\tau| \geq 1 \end{cases}$

(2) $\mu_X(t) = \frac{1}{2}$ 为常数, $R_X(t, t+\tau)$ 只与 τ 有关, 故 $\{X(t), t > 0\}$ 为宽平稳过程

(3) $\langle X(t) \rangle = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} X(t) dt$

$= A \cdot \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} [N(t+1) - N(t)] dt = A$

$P(A = \frac{1}{2}) \neq 1$, $\{X(t), t > 0\}$ 的均值不具有各态历经性

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T16. $X_1, X_2, \cdots$ 独立 $E(X_i)=\mu$, $D(X_i)=\sigma^2>0$

$Y_n = X_n X_{n+1} X_{n+2}$

(1) $\mu_Y(n) = E[Y_n] = \mu^3$

$R_Y(n, n+k) = E[Y_n Y_{n+k}]$

$= E[X_n X_{n+1} X_{n+2} X_{n+k} X_{n+k+1} X_{n+k+2}]$

$|k|\geq 3$ 时, $R_Y(n, n+k) = \mu^6$

$|k|=2$ 时, $R_Y(n, n+k) = \mu^4(\mu^2+\sigma^2)$

$|k|=1$ 时, $R_Y(n, n+k) = \mu^2(\mu^2+\sigma^2)^2$

$|k|=0$ 时, $R_Y(n, n+k) = (\mu^2+\sigma^2)^3$

$\mu_Y(n) = \mu^3$ 为常数, $R_Y(n, n+k)$ 仅与 k 有关

故它为平稳过程

(2) $C_Y(n, n+k) = R_Y(n, n+k) - \mu_Y(n)\mu_Y(n+k)$

$$= \begin{cases} 0 & |k|\geq 3 \\ \mu^{2|k|}(\mu^2+\sigma^2)^{3-|k|} - \mu^6 & |k|\leq 2 \end{cases}$$

有 $\lim_{N\to\infty} \frac{1}{N}\sum_{n=0}^{N} C_Y(n) = 0$ 故均值具有各态历经性

$\langle Y_n \rangle = \mu_Y(n) = \mu^3$

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